Q1. JEE Main 2026 (21 January Shift 2)

Aqueous HCl reacts with MnO_2(s) to form $MnCl_2(aq)$, Cl_2(g) and $H_2O(l)$. What is the weight (in g) of Cl_2 liberated when 8.7 g of MnO_2(s) is reacted with excess aqueous HCl solution?

(Given Molar mass in $gmol^{-1}$ $Mn = 55, Cl = 35.5, O = 16, H = 1$)

(1) 21.3 (2) 71 (3) 14.2 (4) 7.1

Q2. JEE Main 2026 (28 January Shift 2)

For the given reaction:

$CaCO_3 + 2HCl \longrightarrow CaCl_2 + H_2O + CO_2$

If 90 $gCaCO_3$ is added to 300 mL of HCl which contains 38.55%HCl by mass and has density 1.13 g mL^{-1}, then which of the following option is correct ?

Given molar mass of H, Cl, Ca and O are 1, 35.5, 40 and 16 g mol^{-1} respectively.

(1) 64.97 g of HCl remains unreacted

(2) 60.32 g of HCl remains unreacted

(3) 97.30 g of HCl reacted

(4) 32.85 g of $CaCO_3$ remains unreacted

Q3. JEE Main 2026 (22 January Shift 2)

$A + 2B \longrightarrow AB_2$

36.0 g of 'A' (Molar mass: 60 g mol^{-1}) and 56.0 g of ' B′ (Molar mass: 80 g mol^{-1}) are allowed to react. Which of the following statements are correct ?

A. 'A' is the limiting reagent.

B. 77.0 g of AB_2 is formed.

C. Molar mass of AB_2 is 140 g mol^{-1}.

D. 15.0 g of A is left unreacted after the completion of reaction.

Choose the correct answer from the options given below :

(1) C and D Only (2) A and B Only

(3) A and C Only (4) B and D Only

Q4. JEE Main 2026 (22 January Shift 1)

In the reaction,

$2Al(s) + 6HCl(aq) \rightarrow 2Al^{3+}(aq) + 6Cl^{-}(aq) + 3H_2$(g)

(1) 11.2 LH_2(g) at STP is produced for every mole of HCl consumed.

(2) 33.6 LH_2(g) is produced regardless of temperature and pressure for every mole of Al that reacts.

(3) 12 LHCl(aq) is consumed for every 6 LH_2(g) produced.

(4) 67.2 LH_2(g) at STP is produced for every mole of Al that reacts.

Q5. JEE Main 2026 (21 January Shift 1)

14.0 g of calcium metal is allowed to react with excess HCl at 1.0 atm pressure and 273 K. Which of the following statements is incorrect?

[Given : Molar mass in $gmol^{-1}$ of $Ca - 40, Cl - 35.5, H - 1$]

(1) 0.35 mol of H_2 gas is evolved.

(2) 7.84 L of H_2 gas is evolved.

(3) The limiting reagent is calcium metal.

(4) 33.3 g of $CaCl_2$ is produced.

Q6. JEE Main 2026 (21 January Shift 1)

80 mL of a hydrocarbon on mixing with 264 mL of oxygen in a closed U-tube undergoes complete combustion. The residual gases after cooling to 273 K occupy 224 mL. When the system is treated with KOH solution, the volume decreases to 64 mL. The formula of the hydrocarbon is :

(1) C_2H_2 (2) C_2H_4 (3) C_2H_6 (4) C_4H_{10}

ANSWERS AND SOLUTIONS

1. (4) **2.** (1) **3.** (4) **4.** (1) **5.** (4) **6.** (1)

1. **(4)** The balanced redox equation is: $MnO_2 + 4HCl \rightarrow MnCl_2 + Cl_2 + 2H_2O$.
Molar mass of MnO_2 = 55 + 32 = 87 g/mol.
Moles of MnO_2 = $\frac{8.7}{87} = 0.1$ mol.
From stoichiometry, 1 mol MnO_2 produces 1 mol Cl_2, so 0.1 mol produces 0.1 mol Cl_2.
Mass of Cl_2 = $0.1 \times 71 = 7.1$ g.

2. **(1)** Molar masses: HCl = 36.5, $CaCO_3$ = 100 g/mol
HCl solution: mass = 300 × 1.13 = 339 g
HCl present: 339 × 0.3855 = 130.74 g = 130.74/36.5 = 3.58 mol
$CaCO_3$: 90/100 = 0.9 mol
Reaction: $CaCO_3 + 2HCl \rightarrow CaCl_2 + H_2O + CO_2$
HCl required: 0.9 × 2 = 1.8 mol
HCl available: 3.58 mol > 1.8 mol
$CaCO_3$ is limiting. HCl remaining: (3.58 - 1.8) × 36.5 = 1.78 × 36.5 = 64.97 g

3. **(4)** Reaction: A + 2B → AB_2.
Moles of A = 36.0/60 = 0.6 mol;
Moles of B = 56.0/80 = 0.7 mol.
A requires 1.2 mol of B but only 0.7 available, so B is limiting reagent.
So, (A) False - B is limiting.
(B) AB_2 formed = 0.7/2 = 0.35 mol;
Molar mass AB_2 = 60 + 2(80) = 220 g/mol;
Mass = 0.35 × 220 = 77 g True.
(C) False - Molar mass is 220 g/mol.
(D) A remaining = 0.6 - 0.35 = 0.25 mol = 15 g True.
Correct statements: B and D.

4. **(1)** Reaction: $2Al(s) + 6HCl(aq) \rightarrow 2Al^{3+}(aq) + 6Cl^{-}(aq) + 3H_2(g)$
Checking each option:
(1) Per 1 mole HCl: From stoichiometry, 6 mol HCl produces 3 mol H_2, so 1 mol HCl produces 0.5 mol H_2 = 11.2 L at STP. TRUE
(2) Per 1 mole Al: 2 mol Al produces 3 mol H_2, so 1 mol Al produces 1.5 mol H_2 = 33.6 L at STP only, not regardless of T and P. FALSE
(3) Volume ratio at same T,P: 6 mol HCl : 3 mol H_2 = 2:1, so 12 L HCl : 6 L H_2. TRUE
(4) Per 1 mole Al: 1.5 mol H_2 = 33.6 L, not 67.2 L. FALSE
Option (1) is correct.

5. **(4)** Reaction: $Ca + 2HCl \rightarrow CaCl_2 + H_2$

Moles of Ca = $\frac{14.0}{40} = 0.35$ mol. Since HCl is in excess, Ca is limiting.

Option (1): H_2 evolved = 0.35 mol

Option (2): Volume of H_2 at STP = $0.35 \times 22.4 = 7.84$ L

Option (3): Ca is limiting reagent

Option (4): Mass of $CaCl_2 = 0.35 \times 111 = 38.85$ g, not 33.3 g

Statement (4) is incorrect.

6. **(1)** KOH absorbs CO_2. Volume of CO_2 produced = $224 - 64 = 160$ mL

Unreacted $O_2 = 64$ mL, so O_2 consumed = $264 - 64 = 200$ mL

For hydrocarbon C_xH_y: $\frac{CO_2}{\text{Hydrocarbon}} = \frac{160}{80} = 2 \Rightarrow x = 2$

O_2 consumed per mole = $x + \frac{y}{4} = \frac{200}{80} = 2.5$

$2 + \frac{y}{4} = 2.5 \Rightarrow y = 2$

Hydrocarbon is C_2H_2